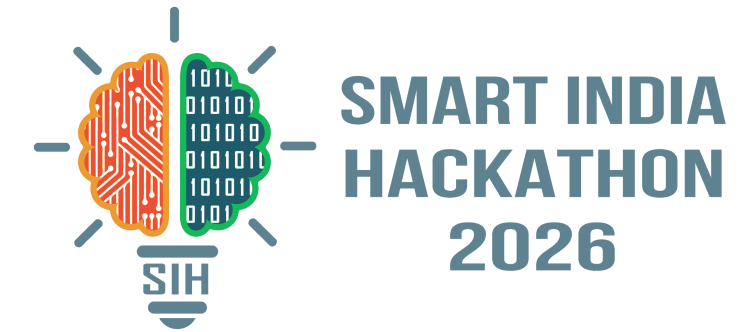
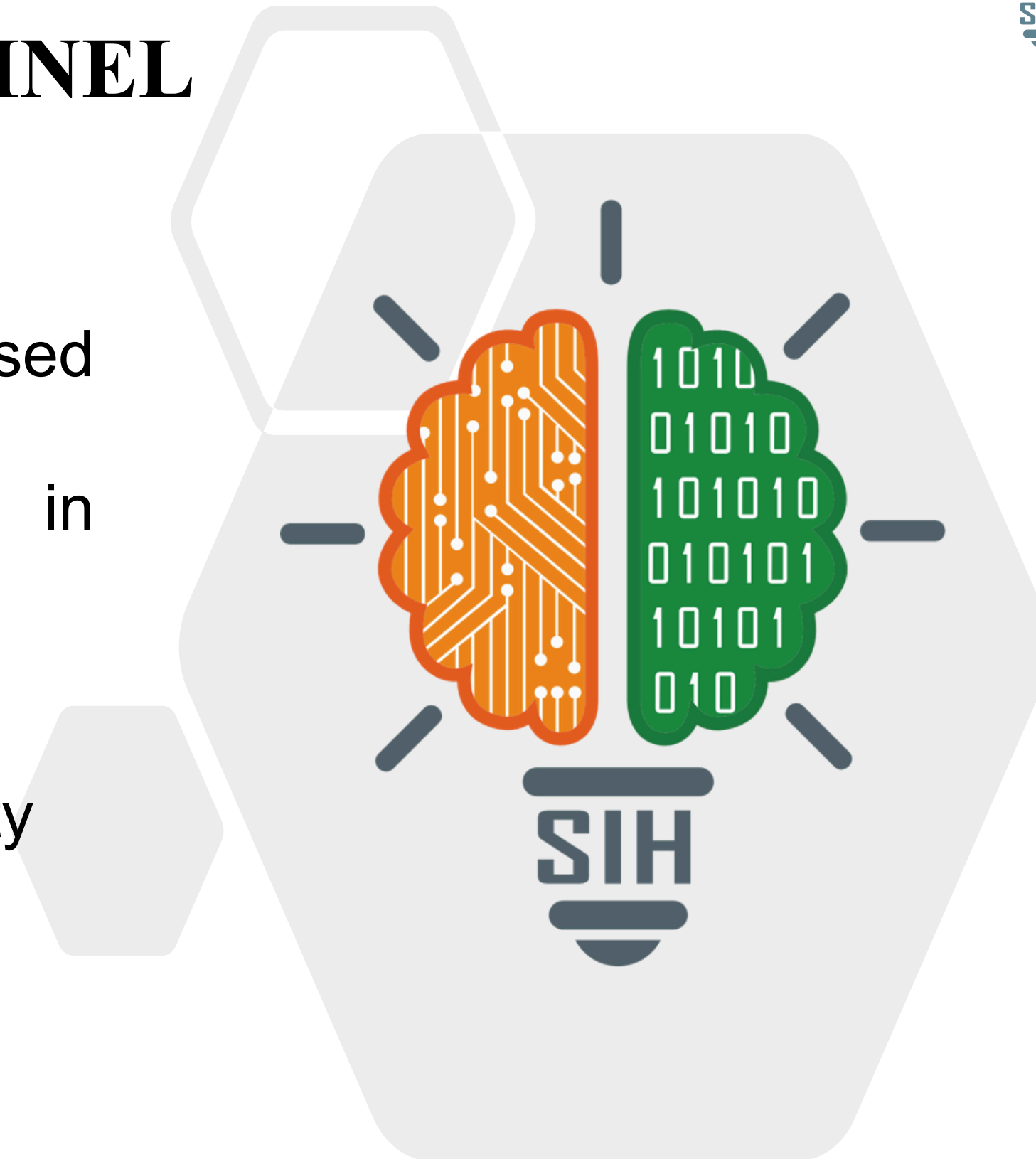


SMART INDIA HACKATHON 2026



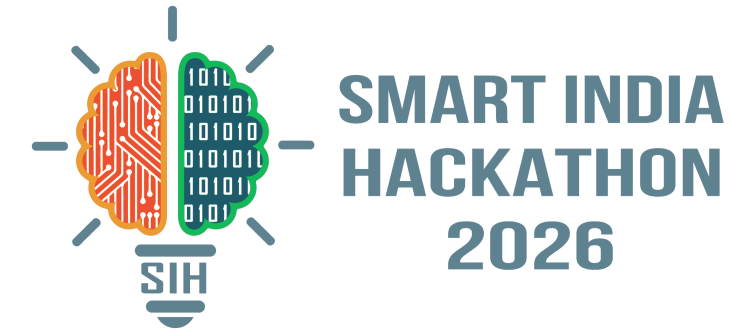
SENTINEL

- **Problem Statement ID** – 26145
- **Problem Statement Title** – AI Based Detection of Cyber Threats in Unidirectional IP Traffic
- **Theme** – Blockchain & CyberSecurity
- **PS Category**- Software
- **Team ID**-
- **Team Name** – Brainy Brews





Sentinel : A passive, hybrid read only threat-detection prototype solution for anomaly detection.



Proposed Solution

- **Passive & Diode-Compliant**
Read-only monitoring of unidirectional traffic with zero return-path communication or active probing.
- **Hybrid ML Detection**
Isolation Forest detects anomalies, followed by 7 XGBoost specialists for precise attack classification
- **Temporal Threat Detection**
A GRU-based temporal engine analyzes multiple flow windows to detect slow and evolving APT attacks.
- **Secure & Explainable Alerts**
SHA-256 blockchain logging ensures tamper-proof alerts, while plain-English explanations show why each alert fired.
- **Multi-Threat Detection**
Detects DDoS, Botnet C2, DGA, DNS Tunneling, Port Scanning, Encrypted Malware, and Data Exfiltration using specialized models.

Prototype Link

- web link
- GitHub Link
- video link

How it Addresses the Problem



Correlates signals into high-confidence alerts.



Identifies novel and zero-day anomalies.



Connects events to detect evolving attacks.



SHA-256 protects alert records from tampering.

Proposed Innovations



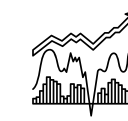
Multi-Model Specialists

7 specialized XGBoost models detect distinct attack types instead of one monolithic classifier.



Temporal Threat Correlation

GRU tracks sequential network behavior to detect low-and-slow and evolving attacks.



Passive & Diode-Compliant

Read-only traffic analysis with zero return-path communication or active network probing.



Explainable & Tamper-Proof

Plain-English alert reasoning combined with an immutable SHA-256 forensic ledger.

Demo Procedure – Attack Kill-Chain Validation

UPLOAD DATA

sentinel_dataset.csv

RECON DETECTED

High ports_per_sec →
Port Scan

C2 DETECTED

Low cv_iat → C2
Beaconing

EXFIL DETECTED

High outbound bytes →
Exfiltration

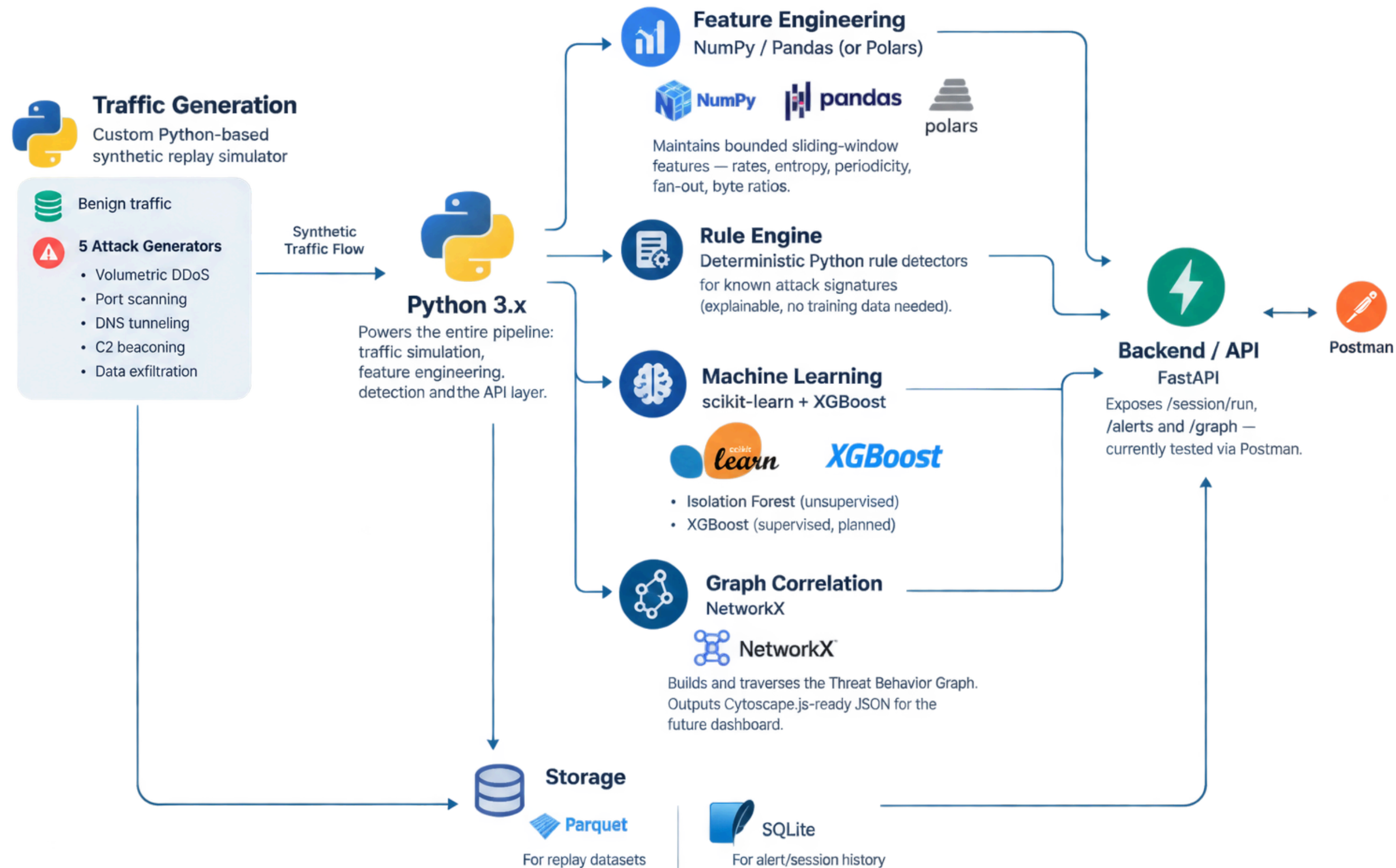
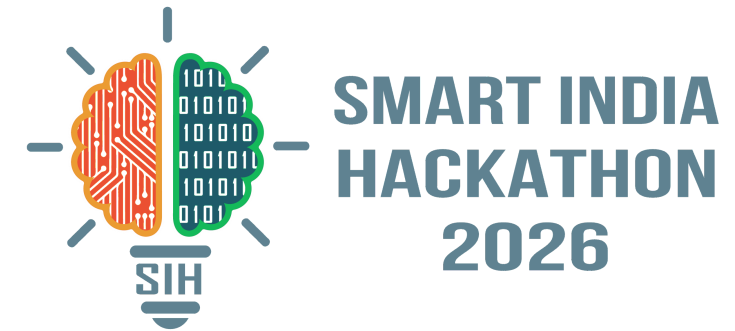
CAMPAIGN
CORRELATED

Campaign Graph → Attack
Chain



TECHNICAL APPROACH

Passive Telemetry + Hybrid ML + Temporal Correlation + Explainable Detection



WHY EACH MODEL?

Isolation Forest

Detects unusual network behavior without labeled attacks.

XGBoost Specialists

Fast and accurate for structured network-flow data.

GRU

Captures patterns across multiple time windows.

Robust Rolling Baselines

Handles outliers and changing host behavior.

NetworkX Evidence Graph

Connects related hosts, IPs and attack events.

Fusion & Anti-Argmax Engine

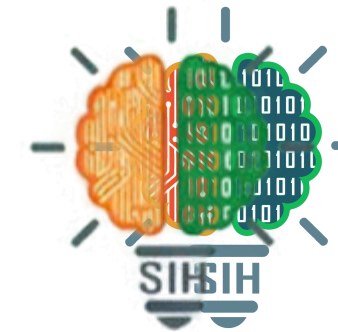
Combines multiple evidence sources instead of trusting one model.

SENTINEL DECISION

ML Score + Baseline + Temporal + Graph + Statistical Evidence → Final Threat Assessment

FEASIBILITY AND VIABILITY

Passive • Offline • Modular • Scalable



1 CAN WE BUILT IT ? YES — Lightweight, Modular & Ready



Open-Source Stack

FastAPI, XGBoost, React & Scikit-Learn



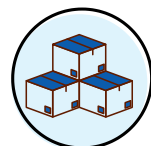
Working Prototype

End-to-end pipeline already operational



Low Compute

Runs on commodity CPUs without GPUs



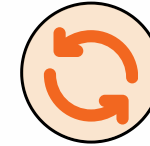
Modular Design

Easily add new models and detectors

FEASIBILITY & VIABILITY

From Prototype to Real-World Impact

2 HOW DO WE SOLVE IT? Robust Pipeline for Real-World Conditions



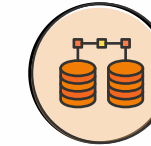
Adaptive Baselines

EWMA-based profiles handle behavior drift



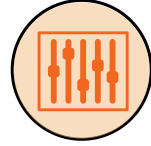
Fast Processing

Vectorized feature extraction reduces bottlenecks



Evidence Fusion

Multiple signals prevent single-feature evasion



Warm-Up Calibration

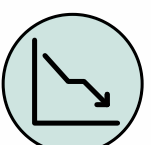
Prevents false alerts on new hosts

3 WHAT CAN GO WRONG? Known Challenges + Mitigations



Model Drift

Continuous retraining adapts to changing traffic



Packet Drops

Optimized flow aggregation handles high throughput



Feature Mimicry

Multi-evidence fusion strengthens detection



Ledger Growth

Merkle checkpoints enable secure archival



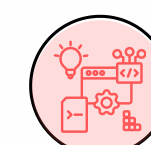
Air-Gap Compatible

Fully passive and diode-compliant



Encryption Agnostic

Detects threats without payload decryption



SIEM/SOAR Ready

Structured alerts integrate with existing tools



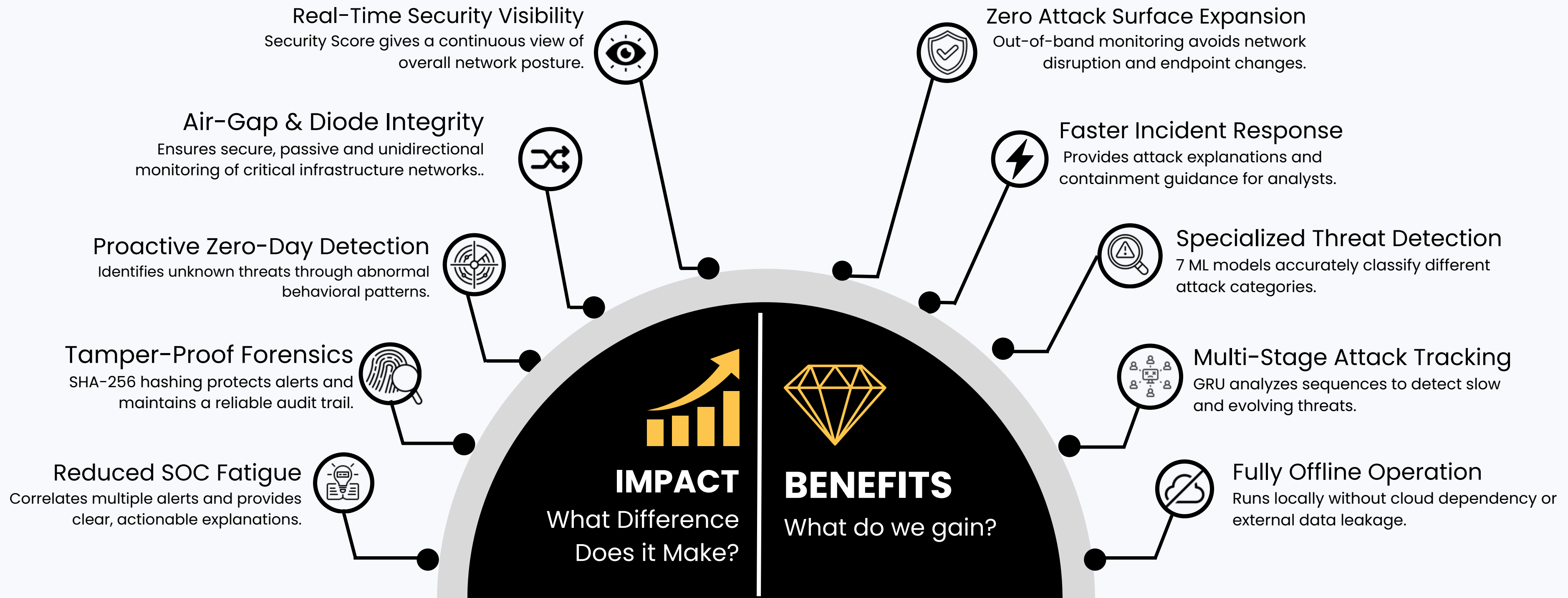
Commodity Hardware

Deployable on standard enterprise servers

4 Can It Operate On-Board? Yes — Offline, Scalable & Mission-Ready

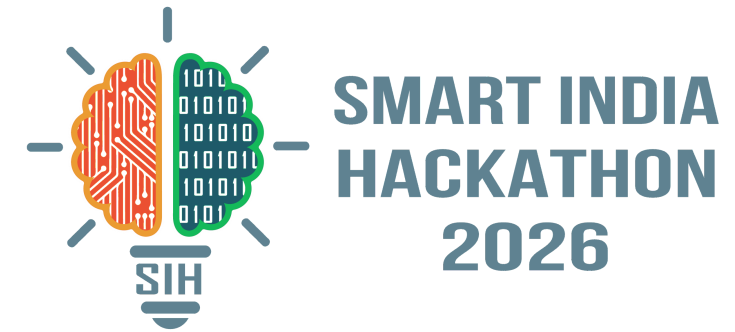
IMPACT AND BENEFITS

Making Scientific Experiments Safer and More Reliable





RESEARCH AND REFERENCES



RELATED SYSTEMS / PRIOR ART

Zeek (formerly Bro) / Suricata

De-facto standards for passive metadata extraction and signature-based packet matching. Heavy reliance on static signatures and deep packet inspection (DPI) causes blindness over encrypted TLS 1.3/QUIC traffic and high alert volume.

zeek.org suricata.io <https://github.com/zeek/zeek>

Darktrace Cyber AI (Enterprise Immune System)

Uses recursive Bayesian clustering to learn "patterns of life" across enterprise devices. Propensity for false positives during routine operational shifts, relies on opaque "black box" anomaly scores, and lacks native cryptographic tamper-proof alert chains.

darktrace.com/cyber-ai-analyst · patents.google.com/patent/US10469509B2

Owl Cyber Defense / Advenica Hardware Data Diodes

Enforces one-way physical layer (fiber-optic TX/RX isolation) data transfer into critical OT/SCADA enclaves. Guarantees zero return path, but provides only physical-layer isolation with no integrated threat forensics or behavioral anomaly intelligence.

owlcyberdefense.com advenica.com/data-diodes

AI / TECHNICAL REFERENCES

1. Isolation Forest — Unsupervised Anomaly Isolation

ieeexplore.ieee.org/document/4781136

2. XGBoost — Scalable Gradient Tree Boosting

arxiv.org/abs/1603.02754

3. GRU (Gated Recurrent Unit) — Sequential Threat Escalation

arxiv.org/abs/1406.1078

4. Cryptographic SHA-256 Hash Chains & Audit Ledgers

merkle.com/papers/Thesis1979.pdf

5. Robust Statistics (Median & MAD) & CUSUM Baselines

biomet.oxfordjournals.org/content/41/1-2/100

6. Explainable AI (XAI) & Grounded Feature Attributions

papers.nips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions

POSITIONING — COMPLEMENTARY TO EXISTING INFRASTRUCTURE DEFENSE, NOT A REPLACEMENT

Existing Systems (Zeek / Snort / Splunk / Inline Firewalls)

Relies on deep packet inspection (DPI) which breaks under TLS 1.3/QUIC encryption; sends active TCP resets/probes that violate hardware diode isolation; generates unweighted alert fatigue with no tamper-proof forensic chain-of-custody.

SENTINEL — Our Technical Contribution

A zero-intrusiveness out-of-band analytics platform providing: **unsupervised benign baseline gating (IF) + 7 supervised XGBoost threat specialists + GRU temporal attack progression + SHA-256 blockchain alert custody + plain-English root-cause explainability** — running strictly passive and offline behind hardware data diodes.